

1 : Import Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score
```

2 : Load Dataset

```
data = pd.read_csv("Insurance_claims.csv")
```

3 : Explore Dataset

```
print(data.head())
```

```
   policy_id  subscription_length  vehicle_age  customer_age  region_code \
0  POL045360                9.3         1.2         41         C8
1  POL016745                8.2         1.8         35         C2
2  POL007194                9.5         0.2         44         C8
3  POL018146                5.2         0.4         44        C10
4  POL049011               10.1         1.0         56        C13

   region_density  segment  model  fuel_type  max_torque  ...  is_brake_assist \
0           8794      C2     M4    Diesel  250Nm@2750rpm  ...           Yes
1           27003      C1     M9    Diesel  200Nm@1750rpm  ...           No
2           8794      C2     M4    Diesel  250Nm@2750rpm  ...           Yes
3           73430      A      M1     CNG   60Nm@3500rpm   ...           No
4           5410     B2     M5    Diesel  200Nm@3000rpm   ...           No

   is_power_door_locks  is_central_locking  is_power_steering \
0                Yes                Yes                Yes
1                Yes                Yes                Yes
2                Yes                Yes                Yes
3                No                 No                 Yes
4                Yes                Yes                Yes

   is_driver_seat_height_adjustable  is_day_night_rear_view_mirror  is_ecw \
0                Yes                No                Yes
1                Yes                Yes                Yes
2                Yes                No                Yes
3                No                 No                No
4                No                 No                Yes

   is_speed_alert  ncav_rating  claim_status
0             Yes           3             0
1             Yes           4             0
2             Yes           3             0
3             Yes           0             0
4             Yes           5             0
```

```
[5 rows x 41 columns]
```

```
print(data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 58592 entries, 0 to 58591
Data columns (total 41 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   policy_id                            58592 non-null  object
1   subscription_length                  58592 non-null  float64
2   vehicle_age                         58592 non-null  float64
3   customer_age                        58592 non-null  int64
4   region_code                         58592 non-null  object
5   region_density                      58592 non-null  int64
6   segment                             58592 non-null  object
7   model                               58592 non-null  object
8   fuel_type                           58592 non-null  object
9   max_torque                          58592 non-null  object
10  max_power                           58592 non-null  object
11  engine_type                         58592 non-null  object
12  airbags                             58592 non-null  int64
13  is_esc                              58592 non-null  object
14  is_adjustable_steering              58592 non-null  object
15  is_tpms                             58592 non-null  object
16  is_parking_sensors                  58592 non-null  object
17  is_parking_camera                   58592 non-null  object
18  rear_brakes_type                    58592 non-null  object
19  displacement                        58592 non-null  int64
```

```

20 cylinder                58592 non-null int64
21 transmission_type        58592 non-null object
22 steering_type            58592 non-null object
23 turning_radius           58592 non-null float64
24 length                   58592 non-null int64
25 width                    58592 non-null int64
26 gross_weight             58592 non-null int64
27 is_front_fog_lights      58592 non-null object
28 is_rear_window_wiper     58592 non-null object
29 is_rear_window_washer    58592 non-null object
30 is_rear_window_defogger  58592 non-null object
31 is_brake_assist          58592 non-null object
32 is_power_door_locks      58592 non-null object
33 is_central_locking       58592 non-null object
34 is_power_steering        58592 non-null object
35 is_driver_seat_height_adjustable 58592 non-null object
36 is_day_night_rear_view_mirror 58592 non-null object
37 is_ecw                   58592 non-null object
38 is_speed_alert          58592 non-null object
39 ncap_rating              58592 non-null int64
40 claim_status            58592 non-null int64
dtypes: float64(3), int64(10), object(28)
memory usage: 18.3+ MB
None

```

```
print(data.shape)
```

```
(58592, 41)
```

4 : Data PreProcessing

```
data = pd.get_dummies(data)
```

Double-click (or enter) to edit

5 : Define Features and Target

```

x = data.drop("claim_status", axis = 1)
y = data["claim_status"]

```

6 : Train and Split

```

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)

```

7 : Train Decision Tree Model

```

model = DecisionTreeClassifier()

model.fit(x_train, y_train)

```

8 : Prediction

```
y=pred = model.predict(x_test)
```

9 : Model Evaluation

```

accuracy = accuracy_score(y_test,y_pred)
print("MODEL ACCURACY : ",accuracy)

```

